

1.1 Atomic Structure

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.1 Atomic Structure
Difficulty	Easy

Time allowed: 50

Score: /39

Percentage: /100

Question 1a

Complete Table 1.1 to show the relative charge and mass of the subatomic particles.

Table 1.1

Subatomic particle	Relative Charge	Relative Mass
Proton		1
Neutron		
Electron		$\frac{1}{1836}$

[4 marks]**Question 1b**

Using the Periodic Table, complete Table 1.2 to show the number of protons, neutrons and electrons in each of the given species.

Table 1.2

Species	Number of protons	Number of neutrons	Number of electrons
^{31}P			
$^{24}\text{Na}^+$			
$^{37}\text{Cl}^-$			

[3 marks]

Question 1c

State why the physical properties of isotopes are different.

[2 marks]

Question 1d

Explain why the chemical properties of ^{35}Cl and ^{37}Cl are similar.

[2 marks]

Question 2a

An isotope of element **X** has two more protons and two more neutrons than an atom of ^{41}K . Use the Periodic Table to identify element **X**.

[1 mark]

Question 2b

Give the full electronic configuration of the following species:

K^+

Ti

Co

[3 marks]

Question 2c

Ionisation energy reactions are endothermic processes.

i)
Write the equations for the first ionisation energy of K and the second ionisation energy of Sc

[2]

ii)
Give the full electronic configuration of the Sc^{2+}

[1]

[3 marks]

Question 2d

Table 2.1 shows successive ionisation energies of element **Z** in Period 3.

Table 2.1

	1	2	3	4	5
Ionisation energy (kJ mol^{-1})	786	1576	3232	4356	16091

i)
State why the **first** ionisation energy generally increases from left to right across the Periodic Table.

[2]

ii)
Using the data of successive ionisation energies in table 2.1, state which group of the Periodic Table element **Z** belongs to.

Group

Explanation

[3]

[5 marks]

Question 3a

Aluminium is a metal in Group 13

Complete the electron configuration in Fig. 3.1 for an aluminium atom using box notation.



Fig. 3.1

[1 mark]

Question 3b

Draw the orbital of the 2s orbital on the axis in Fig. 3.2.

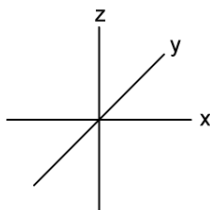


Fig. 3.2

[1 mark]

Question 3c

The first ionisation energy values of Li, Mg and Na are shown in Table 3.1

Table 3.1

Element	First ionisation energy (kJ mol^{-1})
	496
Li	520
	738

Complete the table.

[1 mark]

Question 3d

Explain why the first ionisation energy of aluminium is lower than the first ionisation energy of magnesium.

[2 marks]

Question 4a

Give the electron configuration of the following species:

Mg

Mg^{2+}

[2 marks]

Question 4b

Write an equation including state symbols to show the first ionisation energy of Mg.

[2 marks]

Question 4c

Explain why the second ionisation energy of Mg is higher than the first ionisation energy.

[2 marks]

Question 5a

Fig. 5.1 shows how protons, neutrons and electrons behave differently when they move at the same velocity in an electric field.

Label the positive and negative plates in Fig. 5.1.

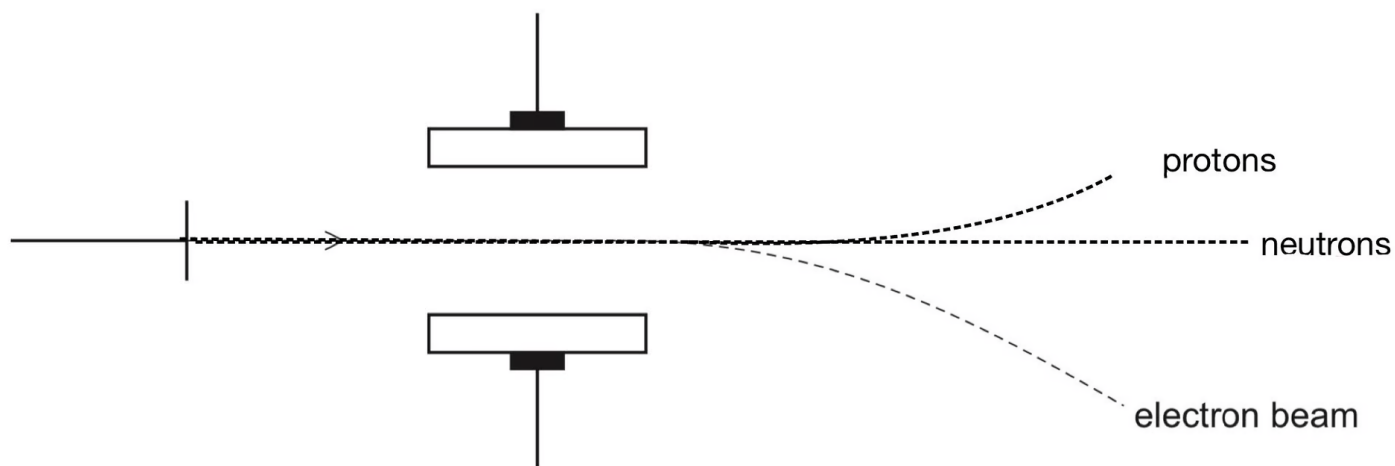


Fig. 5.1

[1 mark]

Question 5b

Using the Periodic Table, complete Table 5.1 to show the number of protons, neutrons and electrons in each of the given species.

Table 5.1

Symbol	Protons	Neutrons	Electrons
^{23}Na			
$^{32}\text{S}^{2-}$			
$^{86}\text{Sr}^{2+}$			

[3 marks]**Question 5c**

Write the electronic configuration for the $^{32}\text{S}^{2-}$ atom.

[1 mark]

Model Answers: Easy

1a

a) The table should be completed as follows:

Subatomic particle	Relative Charge	Relative Mass
Proton	+1; [1 mark]	1
Neutron	0; [1 mark]	1; [1 mark]
Electron	-1; [1 mark]	$\frac{1}{1836}$

[Total: 4 marks]

- These are fundamental basics of this topic so make sure you are confident with this charge and mass table
- Options for the mass of an electron can be:
 - Negligible / very small
 - $\frac{1}{1836}$
 - $\frac{1}{2000}$

1b

b) The completed table should be:

Species	Number of protons	Number of neutrons	Number of electrons
^{31}P	15	16	15
$^{24}\text{Na}^+$	11	13	10
$^{37}\text{Cl}^-$	17	20	18

- All three values correct for ^{31}P ; [1 mark]
- All three values correct for $^{24}\text{Na}^+$; [1 mark]
- All three values correct for $^{37}\text{Cl}^-$; [1 mark]

[Total: 3 marks]

- The number of protons is always equal to the atomic number
 - The number of protons can be found using the atomic numbers on the periodic table
 - The atomic number will be the same for each isotope of the same element
- The number of electrons is equal to the number of protons if the species is neutral
 - If it has a positive charge we must subtract that number of electrons from the number of protons
 - So for $^{24}\text{Na}^+$ we need to use $11 - 1 = 10$ as it is a 1+ charge
 - If it has a negative value we must add that number of electrons to the number of protons
 - So for $^{37}\text{Cl}^-$ we need to use $17 + 1 = 18$ as it has a 1- charge
- The number of neutrons = mass number - atomic number
 - The mass number corresponds to the mass number of the given isotope, not the relative molecular mass M_r

1c

c) The physical properties of isotopes are different because:

- The number of neutrons is different; [1 mark]
- Therefore, the mass number is different; [1 mark]

[Total: 2 marks]

- As a result of different masses, isotopes have different physical properties such as small differences in their mass and density

1d

d) The chemical properties of ^{35}Cl and ^{37}Cl are similar because:

- They are isotopes; [1 mark]
- They have the same number of electrons; [1 mark]

[Total: 2 marks]

- Isotopes of the same element display the same chemical characteristics
- This is because they have the same number of electrons in their outer shells
- Electrons take part in chemical reactions and therefore determine the chemistry of an atom

2a

a) Element **X** is:

- $^{45}_{21}\text{Sc}$ / ^{45}Sc ; [1 mark]

[Total: 1 mark]

- The atomic number of potassium is 19
- Adding two more protons and two more neutrons gives an atomic number of 21 and a mass number of 45
- Scandium has this number of subatomic particles
- Isotopes have a different number of neutrons so you must state the mass number of element **X**
 - This is why just writing Sc will not achieve the mark

2b

b) The electronic configuration of each species is:

- $\text{K}^+ = 1s^2 2s^2 2p^6 3s^2 3p^6$; [1 mark]
- $\text{Ti} = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$; [1 mark]
- $\text{Co} = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$; [1 mark]

[Total: 3 marks]

- When writing a full electronic configuration, you must start with $1s^2$
- Transition metal block elements have an outer electron in the 3d subshell

2c

c)

i) The equations for the first ionisation energy of K and the second ionisation energy of Sc are:

- $\text{K}(\text{g}) \rightarrow \text{K}^+(\text{g}) + \text{e}^-$; [1 mark]
- $\text{Sc}^+(\text{g}) \rightarrow \text{Sc}^{2+}(\text{g}) + \text{e}^-$; [1 mark]

ii) The full electronic configuration of the Sc^{2+} ion is:

- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$; [1 mark]

[Total: 3 marks]

- **Remember:** Ionisation energy only removes 1 mole of electrons at a time
- The second ionisation energy involves removing one mole of electrons from one mole of gaseous ions
- When writing the electronic configuration of elements in the first row of the transition block remember that the 4s subshell empties first
- Sc has an electronic configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$
 - An Sc^{2+} ion will result in an empty 4s shell as electrons will be lost first from this shell

2d

d)

i) Ionisation energies generally increase from left to right across the Periodic Table because:

- The nuclear charge / number of protons / atomic number increases; [1 mark]
- But the shielding / number of shells stays the same; [1 mark]

ii) Element **Z** belongs to:

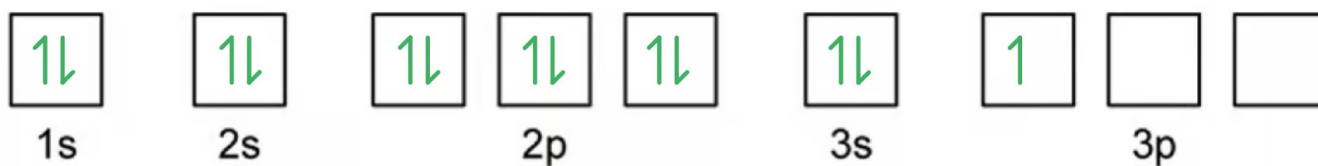
- Group 4 / Group 14; [1 mark]
- (Because,) there is a significant jump from 4th to 5th ionisation energy; [1 mark]
- Moving in an orbit / energy level so electrons are closer to the positive nucleus; [1 mark]

[Total: 5 marks]

- When using successive ionisation data look for a significant jump in ionisation energy
- This is the change from one energy level or orbit closer to the nucleus
- From here you can count backwards to the correct element
- In Table 2.1, the significant change is between the 4th and 5th ionisation energies therefore the element is in Group 4

3a

a) The completed electron configuration for an aluminium atom using box notation is:



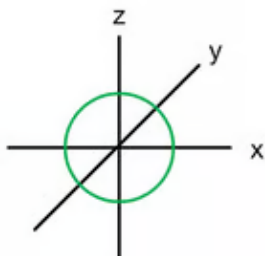
- Correct configuration; [1 mark]

[Total: 1 mark]

- Aluminium is in Group 13 and in Period 3
- The electronic configuration for aluminium is:
 - $1s^2 2s^2 2p^6 3s^2 3p^1$
- Your arrows should be half-headed arrows facing in opposite directions
- It can be helpful to write out the electronic configuration first before writing in the arrows so you have a reference

3b

b) The 2s orbital is:



- Correct shape of s orbital; [1 mark]

[Total: 1 mark]

- s orbitals are spherical in shape
- The s orbitals become larger as the principle quantum number, n , gets larger

3c

c) The completed table is:

Element	First ionisation energy (kJ mol^{-1})
Na	496
Li	520
Mg	738

- Both Na and Mg in the correct place; [1 mark]

[Total: 1 mark]

- Ionisation energy increases from left to right across the Periodic Table and decreases as you go down a group

3d

d) The first ionisation energy of aluminium is lower than magnesium because:

- Aluminium deviates because it's outermost / highest energy / valence electron is in a 3p orbital; [1 mark]

Na	496
Li	520
Mg	738

- Both Na and Mg in the correct place; [1 mark]

[Total: 1 mark]

- Ionisation energy increases from left to right across the Periodic Table and decreases as you go down a group

3d

d) The first ionisation energy of aluminium is lower than magnesium because:

- Aluminium deviates because it's outermost / highest energy / valence electron is in a 3p orbital; [1 mark]
- The 3p orbital is further away from the nucleus / more shielded

OR

3p orbital has a (slightly) higher energy than a 3s orbital; [1 mark]

[Total: 2 marks]

- It is a common exam question to be asked why there are deviations in the trend of ionisation energy across a period
- Remember:** State the orbital that the outer electrons are in to gain the mark
- The same marking points will also apply to Period 2 elements
 - The first ionisation energy increases across the period – for the same reasons
 - Boron has a lower first ionisation energy than beryllium and oxygen has a lower first ionisation energy than nitrogen

4a

a) The electron configuration of:

- Mg = $1s^2 2s^2 2p^6 3s^2$; [1 mark]
- Mg²⁺ = $1s^2 2s^2 2p^6$; [1 mark]

[Total: 2 marks]

- Mg is in Period 3 and in group 2 and contains 12 electrons

- The Mg^{2+} ion has two less electrons than Mg

4b

b) The first ionisation energy of element A



- Correct reactants and products; [1 mark]
- Correct state symbols; [1 mark]

[Total: 2 marks]

- Ionisation energy is the energy required to remove 1 mole of electrons from 1 mole of gaseous atoms / ions
 - The first ionisation energy removes 1 mole of electrons from the atom
 - The second ionisation energy removes 1 mole of electrons from the $1+$ ion
- Be careful to include the state symbols for the gaseous atom and ion

4c

c) The second ionisation energy of element Mg is higher than the first ionisation energy because:

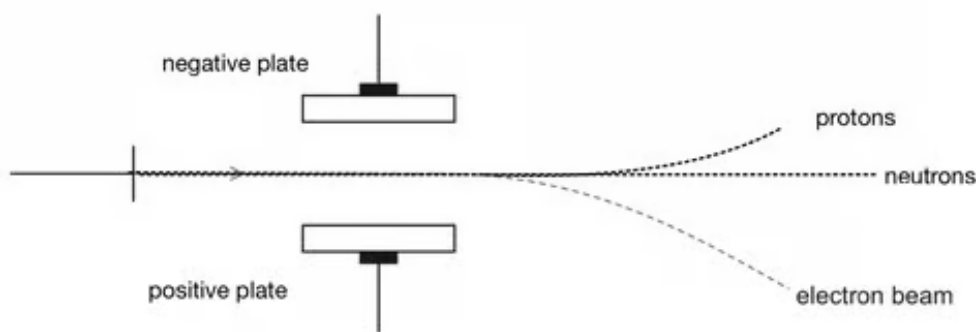
- Electrons being removed from positive ion / $1+$ ion; [1 mark]
- More energy is required to remove 1 mole of electrons from 1 mole of positive ions; [1 mark]

[Total: 2 marks]

- The second ionisation energy is the removal of 1 mole of electrons from 1 mole of gaseous $1+$ ions
- Electrons have a negative charge
- Therefore, there is a strong attraction between the outer electron and the increased ($1+$) positive charge of the nucleus
- This means that more energy is required to remove 1 mole of electrons from the $1+$ ion compared to a neutral atom

5a

a) The correct labelling is:



- Both plates labelled correctly; [1 mark]

[Total: 1 mark]

- Electrons are negatively charged so will be attracted towards the positive plate
- Neutrons are neutral so will pass straight through unaffected
- Protons are positively charged so will be attracted to the negative plate
 - Protons are deflected less than electrons, this also shows that protons are heavier than electrons

5b

b) The completed table should look as follows:

Symbol	Protons	Neutrons	Electrons
^{23}Na	11	12	11
$^{32}\text{S}^{2-}$	16	16	18
$^{86}\text{Sr}^{2+}$	38	48	36

- All three values correct for ^{23}Na ; [1 mark]
- All three values correct for $^{32}\text{S}^{2-}$; [1 mark]
- All three values correct for $^{86}\text{Sr}^{2+}$; [1 mark]

[Total: 3 marks]

- The number of protons can be found using the atomic numbers on the periodic table
- The number of neutrons can be found by using mass number - atomic number

- The number of electrons will be equal to the number of protons
 - Remember when dealing with ions you must account for the charge on the ion when finding the number of electrons
 - For negatively charged ions you need to add the charge to the number of electrons
 - For positively charged ions you need to subtract the charge from the number of electrons

5c

c) The electronic configuration for the $^{32}\text{S}^{2-}$ ion is:

- $1s^2 2s^2 2p^6 3s^2 3p^6$; [1 mark]

[Total: 1 mark]

- A sulfur atom contains 16 electrons
 - Its electronic configuration is $1s^2 2s^2 2p^6 3s^2 3p^4$

